Metropolitan University

Department of Computer Science and Engineering

Summer Semester Mid Term Examination - 2024

Program: B.Sc. in CSE **Batch: 57**

Course: CSE 321: Operating System

Time: 1 Hour **Marks:** 15

Answer any **three** sets of questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

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1. (a) What is a System Call? (1)

   (b) Explain the functionalities of the following system calls: **fork()**, **getpid()**, **pipe()** & **chmod()**. (4)

2. (a) What is a Process? (1)

   (b) Name and describe two IPC mechanisms. (4)

3. Suppose that the following processes arrive for execution at the times indicated. Each process will run for the amount of time listed. In answering the questions, base all decisions on the information you have at the time the decision must be made.

Process	Arrival Time	Burst Time	Priority
P_0	2	1	1
P_1	3	2	2
P_2	1	8	3
P_3	0	4	4
P_4	1	3	0

   (a) What will be the average waiting time for these processes if we use the **Priority** scheduling algorithm? Draw the Gantt Chart and explain. Consider lower value refers to higher priority. (5)

4. Suppose that the following processes arrive for execution at the times indicated. Each process will run for the amount of time listed. In answering the questions, base all decisions on the information you have at the time the decision must be made.

Process	Arrival Time	Burst Time
P_0	2	1
P_1	3	2
P_2	1	8
P_3	0	4
P_4	1	3

   (a) What will be the average waiting time for these processes if we use the **Round Robin** scheduling algorithm with **Time Quanta=3**? Draw the Gantt Chart and explain. (5)

Metropolitan University

Department of Computer Science and Engineering

Summer Semester Final Examination - 2024

Program: B.Sc. in CSE; **Batch: 57**

Course: CSE 321: Operating System

Time: 2 Hours **Marks: 40**

Answer any **four** sets of questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

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1. (a) Indicate whether each of the following statements is true or false: (5)

   1. Virtual memory space is always smaller than physical memory space F
   2. Segmentation avoids external memory fragmentation F
   3. The operating system is not responsible for resource allocation between competing processes. F
   4. Internal fragmentation occurs due to fixed-sized blocks of memory. T
   5. A Trojan Horse is a code segment embedded in the legitimate program. T

   (b) Consider the following Segment table: (5)

Segment	Base	Limit
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

   What are the physical addresses for the following logical addresses? If a logical address exceeds the limit, indicate it as invalid.
   i) 0, 430 ii) 1, 10 iii) 2, 500 iv) 3, 400 v) 4, 112

2. (a) State Amdahl's Law and explain its relevance to parallel processing in operating systems. (2)

   (b) Given a system where 80% of the workload can be parallelized, and the remaining 20% must be executed sequentially, calculate the theoretical speedup achievable using Amdahl's Law when running the workload on a system with 4 processors. (4)

   (c) Justify how the behavior of thread execution changes when moving from a single-core processor to a multi-core processor, and discuss the implications for performance. (4)

Process	Allocation	Max	Available
	ABCD	ABCD	ABCD
P_1	0 0 1 2	3 4 1 2	3 4 1 3
P_2	2 1 0 0	5 7 5 3	
P_3	3 3 2 1	3 5 5 6	
P_4	0 1 4 2	7 6 5 2	
P_5	0 0 1 2	0 6 7 6	

Table 1: System

8 9 9 10

3. (a) Consider the system depicted in Table 1. Calculate the Need Matrix. Illustrate whether the system is in a safe state by demonstrating an order in which the processes may complete, or an unsafe state demonstrating which process cannot complete, using the banker's algorithm. (10)

4. (a) How does a boot sector virus work? Show with a diagram. (5)

   (b) Define: Worm, Kernel View, Logic Bomb, Semaphore, Context Switch. (5)

5. (a) Consider the following resource allocation graph:
   P = { P1, P2, P3, P4 }
   R = { R1, R2, R3 }
   E = { R1→P1, P1→R2, R2→P2, P2→R3, R3→P3, P3→R1, R1→P4 }
   Resource type R1 has two instances, R2 has one instance and R3 has one instance. Make the resource allocation graph. If deadlock exists, identify the processes and resources involved in the cycle. (5)

   (b) Explain Peterson's Solution with an example of two threads competing for shared data. (5)

6. (a) What is a kernel? (2)

   (b) Analyze the scenarios in which preemptive scheduling and non-preemptive scheduling occur. (4)

   (c) Illustrate the process states with a diagram and provide a description. (4)

Metropolitan University, Sylhet

Department of Computer Science and Engineering
Midterm Examination – Summer, 2023
Program: B.Sc. in CSE; **Batch: 57**
Course: CSE 215: Communication Engineering

Time: 1 Hour 30 Minutes | Marks: 30

Answer any **THREE** questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

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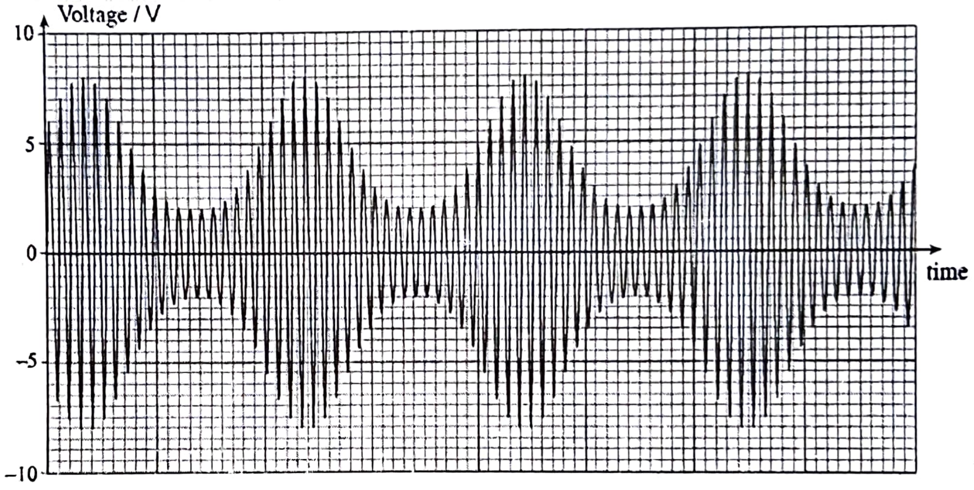
1. a) What is Modulation? Briefly describe the types of modulation when the carrier is an analog signal. 4

   b) Find the percentage of modulation from the following AM signal. Is there any overmodulation observed? Justify your answer. 4+2



2. a) Describe the types/modes of electronic communication systems with appropriate examples. 3

   b) Define the following terms: 3

      i. Attenuation ii. Distortion iii. SNR

   c) Find the cut-off frequency of a Low Pass Filter where, Resistance, R = 4.7 KΩ and Capacitance, C = 560 pF. Draw the circuit diagram of the filter using given components. 4

Or,

   a) Discuss the functions of a local exchange. 3

   b) Suppose 1300 calls are offered to a channel and 8 calls are lost. Duration of a call is 2 minutes. Find: (a) Offered Traffic, *A* (b) Carried traffic (c) Lost traffic (d) GOS, *B* (e) Congestion Time. 2.5

   c) Write the different range and application for radio waves, microwaves and infrared waves. 2.5

   d) Define (i) Antenna gain (ii) Antenna Polarization (iii) Antenna efficiency and (iv) Antenna Directivity with equation. 2

3. a) Explain the basic requirements of complete telecommunication system in short. 3

b) Figure out the one way line and radio telecommunication channel (transmitter end and receiver end) with marking each part. 2

c) Differentiate between digital and analog signal. 3

d) Define bandwidth and Frequency spectrum. 2

4. a) Write the working procedure of any one of the following switching system – 3

    i. Step by step

    or,

    ii. Reed Relay

b) What is noise? Mention different types of noise. 3

c) List different types of topology with figure. 2

d) What is a switching system? Why it is required? 2

5. a) Draw and discuss about SPC block diagram. 4

b) Discuss about delta and pulse modulation. 3

c) Write short notes on (i) ASK (ii) FSK (iii) PSK. 3

Metropolitan University
Department of Computer Science and Engineering
Summer Semester Final Examination – 2023
Program: B.Sc. in CSE; Batch: 57th (All Section)
Course: CSE 215: Communication Engineering

Time: 2 Hours **Marks: 40**

Answer any **four** questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

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1. (a) Figure out GSM Architecture. 3
   (b) What is cell? Discuss multiple Access technique with figure. 1+3
   (c) Write the radio parameter and characteristics of GSM-900. 3

2. (a) Draw General Packet Radio Service (GPRS). 3
   (b) Write the functionalities of BTS and BSS. 4
   (c) Discuss working procedure of SIM with figure. 3

3. (a) Describe call termination steps of GSM with figure. 4
   (b) Write the functionalities of MSC and HLR. 4
   (c) Write short note on CAMEL and HSCSD. 2

4. (a) Write the steps of mobile call origination with figure. 5
   (b) Write the steps of speech encoding with figure. 5

5. (a) Write the steps of mobile inter MSC Handoff mechanism with figure. 5
   (b) Discuss logical channel with figure of GSM-900. 5

6. (a) Write about (i) New supplementary services (ii) New features and Capabilities of GSM. 2
   (b) Discuss protocol model with figure of GSM-900. 5
   (c) Discuss Location Update System with figure of GSM-900. 3

Metropolitan University, Sylhet

Department of Computer Science and Engineering
Summer Semester Final Examination – 2025
Program: B.Sc. in CSE; **Batch:** 57
Course: CSE 421: Artificial Intelligence

Time: 2 Hours **Marks: 40**

Answer any **four** sets of questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

1. a) Explain what happens if **no activation function** is used between layers in a neural network. 2

b) Define encoder and decoder in NLP. 3

c) Define the stages involved in preprocessing text data in NLP workflows. 5

2. a) Define plateau, ridge, and local optima in the context of local search algorithms. 1

b) You are given two jugs, a 7-gallon one and a 4-gallon one, a pump which has unlimited water which you can use to fill the jug, and the ground on which water may be poured. Neither jug has any measuring markings on it. How can you get exactly 5 gallons of water in one of the jugs? 5

c) Translate these sentences into formulas in First Order Logic (Propositional Logic): 4
i) All students in the class study AI.
ii) Some students like mathematics.
iii) There exists a teacher who teaches all students.
iv) CSE125 is a mathematical course.

3. a) Define constraint satisfaction problem. 2

b) Suppose we have a simple cost function: 8

$$y(x) = (x - 5)^2$$

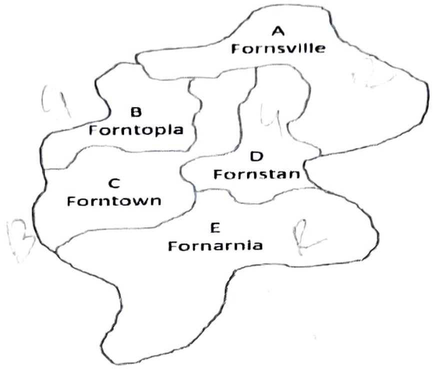
Using **gradient descent** with a learning rate, α = **0.01**, and starting from an initial value of **x** = **0**, and **b** = **0**, compute the first two iterations of gradient descent. Show cost function reduction.

4. a) Define the difference between fully observable and partially observable environments. Give one example of each. 2

b) Give an example of a PEAS description for a self-driving car and explain its components briefly 4

c) Draw the architecture of a model based agent. 4

5. a) Represent this as a Constraint Satisfaction Problem (CSP) by defining the variables, domains, and constraints. Using three colors {Red, Green, Blue}, color the map of **Graph 1** so that no two adjacent regions have the same color. 3



Graph 1

b) The following dataset shows whether a person buys a computer (**Yes/No**) based on **Age**, **Income**, and **Student** status. 7

Person	Age	Income	Student	Buys Computer
1	Youth	High	No	No
2	Youth	Medium	Yes	No
3	Middle	High	No	Yes
4	Youth	Medium	No	Yes
5	Senior	Low	Yes	Yes
6	Middle	Low	Yes	Yes
7	Youth	Medium	Yes	Yes
8	Youth	Medium	Yes	No

Using the **Naive Bayes algorithm**, predict whether a new person with Age = **Youth**, Income = **Medium**, Student = **Yes** will buy a computer.

6. **a)** Define forward propagation & backward propagation. 2

b) *Apply Perceptron Training Algorithm to the following dataset. Use* $\alpha = 0.6, T_h = 1.4$ (if $\sum x_i w_i \geq 1.4$ then output = 1), $w_1 = -0.4$ and $w_2 = 0.5$ for two iterations. Show the final updated weights using the **figure of a perceptron.** 8

X1	X2	Y
0	0	0
0	1	0
1	0	1
1	1	1

Artificial Intelligence

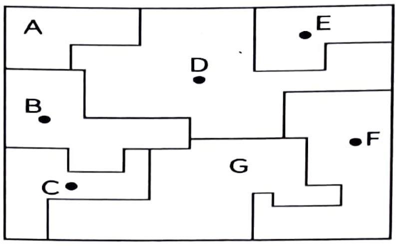
Class Test 3

Batch : 57E

Time: 40 minutes **Marks: 20**

Answer **ALL** questions.

1) Represent this as a Constraint Satisfaction Problem (CSP) by defining the variables, domains, and constraints. Using three colors {Red, Green, Blue}, color the map so that no two adjacent regions have the same color. 5



2) Given the dataset below, use **Naive Bayes classification** to determine whether a new email with features Free=Yes, Win=Yes, Click=No is classified as "Spam" or "Not Spam". 6

Email	Free	Win	Click	Label
1	Yes	Yes	Yes	Spam
2	No	Yes	No	Not Spam
3	Yes	No	Yes	Spam
4	No	Yes	Yes	Not Spam
5	Yes	Yes	No	Spam

3) What is encoder in NLP? 2

4) Consider the following training data: 8

X	Y
0	4
1	7
2	7
3	8

Show the gradient descent steps to optimize the cost function. Where, the initial $h_\theta(x)$ *function*, $\theta_0 = 1$, *and* $\theta_1 = 2$, *Learning rate* $\sigma = 0.001$. Perform 1 iteration to show the process.

Course: CSE 421: Artificial Intelligence

Class Test 2 Set : সমুদ্র

Batch : 57E

Time: 40 minutes **Marks: 20**

Answer **ALL** questions.

1. Why is the **ReLU** function often preferred in deep networks over sigmoid or tanh? (2)
2. Differentiate between uninformed and informed search. (3)
3. Solve the following Puzzle with Heuristic Search (State Space Search)- (5)

S

1	2	3
	4	6
7	5	8

G

1	2	3
4	5	6
7	8	

4. (i) Define State Space Search. (2)

(ii) Apply Perceptron Training Algorithm to the following dataset. Use $\alpha = 0.6$, (8) $T_h = 1.4$ (if $\sum x_i w_i \geq 1.4$ then output = 1), $w_1 = -0.6$ and $w_2 = 0.7$ for two iterations. Show the final updated weights using the **figure of a perceptron.**

X1	X2	Y
0	0	0
0	1	1
1	0	1
1	1	1

Course: CSE 421: Artificial Intelligence

Class Test 2 Set: আকাশ

Batch : 57E

Time: 40 minutes **Marks: 20**

Answer **ALL** questions.

1.Why is the **ReLU** function often preferred in deep networks over sigmoid or tanh? (2)

2. Differentiate between uninformed and informed search. (3)

3. Solve the following Puzzle with Heuristic Search (State Space Search)- (5)

S

2	8	3
1	6	4
7		5

G

1	2	3
4	5	6
7	8	

4. (i) Define Backward Propagation. (2)

(ii) Apply Perceptron Training Algorithm to the following dataset. Use $\alpha = 0.3$, $T_h = 1.3$ (if $\sum x_i w_i \geq 1.3$ then output = 1), $w_1 = -0.5$ and $w_2 = 0.5$ for two iterations. Show the final updated weights using the **figure of a perceptron.** (8)

X1	X2	Y
0	0	0
0	1	1
1	0	1
1	1	1

Course: CSE 421: Artificial Intelligence

Class Test 1

Batch : 57E

ID :

Time: 40 minutes **Marks: 20**

Answer **ALL** questions.

1. Fill the table 6

Task Environment	Observable	Deterministic	Episodic	Agents
Email Spam Filter				
Crossword puzzle				
Taxi Driving				

2. What is the difference between artificial general intelligence (AGI) and narrow AI? Illustrate with real-world examples. 4
3. What is the PEAS description, and why is it important in agent design? 5
4. Draw the architecture of a utility based agent. 5